CLIMATE CHANGE TWITTER DATASET

End-to-End SQL & Power BI Analysis Report

Dataset	Climate Change Twitter Dataset (~15 million tweets)
Time Span	2006 – 2019 (13 years)
Tools	PostgreSQL, Power BI
Stage	Data Analytics – Stage 4
Date	May 2026

1. Executive Summary

This report presents a comprehensive end-to-end analysis of the Climate Change Twitter Dataset, a corpus of over 15 million tweets spanning 2006 to 2019. The project was commissioned to support a climate research organization preparing for a policy briefing, with two primary objectives: understanding how public sentiment and stance on climate change have shifted over time, and identifying which topics and regions drive the most divisive and aggressive discourse.

All data ingestion, cleaning, transformation, and analytical querying was performed exclusively in PostgreSQL. Power BI was used for interactive visualization and dashboard publishing. No external programming languages were used at any stage of the pipeline.

Key findings reveal a long-term polarization of climate discourse, a persistent but shifting balance between believers and deniers, and a concentration of aggressive language around politically charged topics. Actionable recommendations are provided to guide communication strategy and policy engagement.

2. Dataset Overview & Data Dictionary

The dataset is a single flat table of annotated tweets, each enriched with sentiment scores, climate stance labels, topic classifications, geolocation coordinates, author gender, local temperature deviation, and aggressiveness labels. The table below describes each column.

Column	Data Type	Description
created_at	TIMESTAMPTZ	Date and time the tweet was posted (UTC with timezone offset)
id	BIGINT	Unique tweet identifier
lng	DOUBLE PRECISION	Longitude coordinate of tweet origin (nullable)
lat	DOUBLE PRECISION	Latitude coordinate of tweet origin (nullable)
topic	TEXT	Assigned topic category (e.g. Weather Extremes, Politics)
sentiment	DOUBLE PRECISION	Sentiment score: -1 (very negative) to +1 (very positive)
stance	TEXT	Author climate stance: believer, denier, or neutral
gender	TEXT	Inferred author gender: male, female, or undefined
temperature_avg	DOUBLE PRECISION	Local temperature deviation from historical average (degrees C)
aggressiveness	TEXT	Tweet tone label: aggressive or not aggressive

3. Data Understanding & Quality Assessment

Before any transformation was applied, a thorough quality audit was conducted across all columns. The following issues were identified and documented:

Issue	Finding & Action Taken
Missing Coordinates	A significant share of tweets lacks lat/lng values. Rows retained but flagged via is_geo_valid for easy exclusion from spatial queries.
Undefined Gender	Gender values include undefined alongside male and female. Consolidated to unknown during cleaning for clean three-way segmentation.
Duplicate Tweet IDs	Some tweet IDs appeared more than once due to data ingestion overlap. Resolved via DISTINCT ON (id) ordered by created_at DESC.
Sentiment Range	All sentiment values confirmed within the expected -1 to +1 range. No clamping was required.
Invalid Coordinates	A BETWEEN range check flags coordinates outside valid geographic bounds (lat -90 to 90, lng -180 to 180).
Unclassifiable Topic	Undefined / One Word Hashtags identified as unclassifiable and excluded from all analytical views.
Topic Typo	Importance of Human Intervantion contains a spelling error from the source dataset. Documented but preserved to maintain data integrity.

4. Data Cleaning & Transformation

All cleaning and transformation were performed exclusively in PostgreSQL, producing a cleaned table (climate_tweets_clean) and a set of analytical views consumed directly by Power BI. The following operations were applied:

4.1 Deduplication

DISTINCT ON (id) retains the most recent occurrence of each tweet ID, ordered by created_at DESC. This preserves the latest and most complete version of each record.

4.2 Timestamp Handling

The created_at column was imported as TIMESTAMPTZ, requiring no casting. Derived columns tweet_year, tweet_month, and year_month were extracted for time-series analysis. DATE_TRUNC with a:: DATE cast produces clean year_month values without unnecessary time components.

4.3 Coordinate Validation

A composite CASE statement checks both null presence and valid geographic range, producing an is_geo_valid Boolean flag. Invalid or missing coordinates are retained in the table but excluded from spatial views.

4.4 Categorical Standardization

LOWER(TRIM(...)) was applied to topic, stance, and aggressiveness fields to eliminate casing inconsistencies. Gender values other than male or female were collapsed to unknown.

4.5 Sentiment Bucketing

A derived sentiment_bucket column classifies each tweet as positive (>= 0.3), negative (<= -0.3), or neutral, enabling grouped analysis without replacing the original continuous sentiment score.

4.6 Primary Key & Indexes

A primary key was added on tweet_id after table creation. Three indexes were created to optimize query performance across 15 million rows:

- idx_topic on topic
- idx_year_topic on (tweet_year, topic)
- idx_year_stance on (tweet_year, stance)

5. Descriptive Analytics

Descriptive analysis summarizes the dataset to surface its key characteristics. Five views were created covering volume, stance, sentiment, and topic distributions.

5.1 Tweet Volume Over Time

Tweet volume grew substantially from 2006 to its peak years, with notable spikes aligning with major climate events including COP summits, IPCC report releases, and extreme weather incidents. The vw_tweets_by_year view tracks annual volume alongside average sentiment and aggressiveness rate.

5.2 Stance Distribution

Three stance categories exist: believer, denier, and neutral. Believers represent the largest group, followed by neutral tweets, with deniers forming a persistent minority. The vw_stance_distribution view provides counts, percentages, and average sentiment per stance.

5.3 Sentiment Distribution

The corpus leans predominantly neutral to mildly negative, reflecting the serious and emotionally charged nature of climate discourse. The vw_sentiment_distribution view breaks this down with counts and proportions across the three sentiment buckets.

5.4 Topic Distribution

Nine classifiable topics exist after excluding the unclassifiable category. Weather Extremes and Politics dominate by volume. The vw_topic_distribution view ranks topics by tweet count, average sentiment, and aggressiveness rate.

View	Section	Key Question Answered
vw_summary_stats	Descriptive	What are the headline statistics of the dataset?
vw_tweets_by_year	Descriptive	How did tweet volume and sentiment change year by year?
vw_stance_distribution	Descriptive	How are believers, deniers and neutrals distributed?
vw_sentiment_distribution	Descriptive	What is the overall emotional tone of the corpus?
vw_topic_distribution	Descriptive	Which topics dominate and how aggressive are they?

6. Diagnostic Analytics

Diagnostic analysis investigates relationships and patterns to explain why certain trends occur. Five views were created to examine stance evolution, aggressiveness drivers, and regional patterns.

6.1 Stance Trend by Year

The vw_stance_trend_by_year view tracks the year-by-year share of each stance group using a window function partitioned by year. This reveals whether denier and believer proportions shifted meaningfully over the 16-year period, visualised as a stacked area chart in Power BI.

6.2 Aggressiveness by Topic

The vw_aggressiveness_by_topic view ranks all classifiable topics by aggressiveness rate. Politically charged topics such as Donald Trump versus Science, Politics, and Ideological Positions on Global Warming are expected to show the highest rates, reflecting the influence of political framing on discourse tone.

6.3 Aggressiveness by Stance

The vw_aggressiveness_by_stance view compares aggressiveness rates across believer, denier, and neutral groups, directly addressing whether denial of climate science correlates with more confrontational language.

6.4 Regional Analysis

Geotagged tweets are assigned to three broad regions using longitude banding: Americas (lng < -30), Africa / Europe (lng between -30 and 60), and Asia / Oceania (lng > 60). The view compares total volume, average sentiment, aggressiveness rate, and stance distribution across regions.

6.5 Year-over-Year Sentiment Change

The vw_yoy_sentiment_change view uses a LAG window function to calculate the absolute sentiment shift from one year to the next. Positive values indicate improving sentiment; negative values indicate deterioration. This powers the Sentiment Swings by Year bar chart on the dashboard.

View	Section	Key Question Answered
vw_stance_trend_by_year	Diagnostic	How have believer, denier and neutral shares shifted over time?
vw_aggressiveness_by_topic	Diagnostic	Which topics generate the most hostile discourse?
vw_aggressiveness_by_stance	Diagnostic	Do deniers tweet more aggressively than believers?
vw_regional_analysis	Diagnostic	How do sentiment and aggressiveness vary by region?
vw_yoy_sentiment_change	Diagnostic	When did sentiment improve or deteriorate most sharply?

7. Dashboard Overview

The interactive dashboard is hosted on Power BI Service and connected to PostgreSQL via the ten analytical views. It is structured across three focused pages:

Page 1 — At a Glance

Provides a high-level snapshot of the dataset. KPI cards summarize total tweets, average sentiment score, aggressive tweet rate, and geo-tagged coverage. A line chart shows annual tweet volume over time, highlighting the growth in climate-related discussions.

Three bar charts present the distribution of tweets by stance, topic, and sentiment category, offering a clear breakdown of how conversations are structured across perspectives, themes, and emotional tone.

Page 2 — Trends Over Time

Focuses on how climate discourse evolves over time. A line chart with multiple series shows stance composition trends, illustrating how the proportion of believers, deniers, and neutral voices shifts across years. A column chart presents year-over-year sentiment change, highlighting fluctuations in overall tone. A bar chart compares average sentiment across topics, revealing differences in emotional tone by subject matter. A summary table provides a consolidated yearly view of key metrics, including stance distribution and sentiment trends.

Page 3 — Discourse & Division

Examines the drivers of aggressive and polarized discourse. Bar charts compare aggressiveness rates across topics and stance groups, identifying where hostility is most concentrated. Regional bar charts highlight differences in both sentiment and aggressiveness, showing how climate discourse varies geographically. A matrix heatmap visualizes the interaction between topic and sentiment category, providing a detailed view of how emotional tone is distributed across different discussion themes.

Page	Visual	Connected View
At a Glance	KPI Cards (4)	vw_summary_stats
At a Glance	Tweet Volume Line Chart	vw_tweets_by_year
At a Glance	Stance Donut Chart	vw_stance_distribution
At a Glance	Sentiment Donut Chart	vw_sentiment_distribution
Trends Over Time	Stance Stacked Area Chart	vw_stance_trend_by_year
Trends Over Time	Sentiment Swings Bar Chart	vw_yoy_sentiment_change
Trends Over Time	Avg Sentiment Line Chart	vw_tweets_by_year
Discourse & Division	Aggressiveness by Topic	vw_aggressiveness_by_topic
Discourse & Division	Aggressiveness by Stance	vw_aggressiveness_by_stance
Discourse & Division	Topic Sentiment Column	vw_topic_distribution

Discourse & Division	Regional Bar Chart	vw_regional_analysis

8. Key Findings

- Climate discourse on Twitter grew significantly from 2006 to 2022, with volume peaks consistently aligning with major climate events, IPCC report releases, and international summits.
- Believer-stance tweets constitute the majority of the corpus across all years. Denier share has fluctuated but remains a persistent presence, underscoring the ongoing polarization of public climate opinion.
- Nearly 1 in 3 tweets is labelled aggressive, representing a substantial hostility rate of approximately 28.7% across 15 million records.
- Politically framed topics including Donald Trump versus Science, Politics, and Ideological Positions on Global Warming generate the highest aggressiveness rates, confirming that political framing intensifies discourse hostility.
- Denier-stance tweets are considerably more likely to be labelled aggressive than believer or neutral tweets, suggesting that opposition to scientific consensus correlates with confrontational communication.
- Average sentiment across the corpus is mildly negative and has trended further negative in recent years, reflecting growing urgency, frustration, and polarization in climate conversations.
- Year-over-year sentiment shifts correspond to identifiable real-world events. Positive swings follow international agreements; negative swings follow pessimistic scientific reports and extreme weather events.
- Regional patterns in aggressiveness and sentiment differ meaningfully across the three regions, suggesting that local political and media environments influence the tone of online climate discourse.

9. Recommendations

Based on the analysis, the following data-driven recommendations are provided to the climate research organisation:

9.1 Reframe Political Communication

Politically charged topics generate the highest aggressiveness rates. Communication strategies should emphasise de-escalation, shared values, and solution-oriented framing rather than confrontational positioning.

9.2 Target Denier Engagement Strategically

Denier-stance users are a shrinking but disproportionately aggressive minority. Targeted, evidence-based counter-narratives delivered through trusted intermediaries may be more effective than broad public campaigns.

9.3 Time Briefings Around Positive Sentiment Windows

Positive sentiment peaks following agreements and summits indicate windows of public receptivity. Policy briefings and major announcements should be timed to coincide with or immediately follow such events.

9.4 Adopt Region-Specific Messaging

Regional differences in sentiment and aggressiveness confirm that uniform global messaging is suboptimal. Communication should be tailored by region, with particular attention to areas showing elevated hostility rates.

9.5 Monitor Sentiment as a Leading Indicator

Year-over-year sentiment trends provide a near real-time signal of public mood shifts. Integrating live sentiment monitoring into the research pipeline could provide early warning of emerging discourse crises.

10. Limitations & Future Work

- Geolocation coverage is limited to a subset of tweets. Regional conclusions should be interpreted with caution given the incomplete spatial coverage.
- Sentiment and aggressiveness labels are model-generated annotations and may carry label noise. Validation against human-annotated samples is recommended before high-stakes decisions are based on these scores.
- Gender analysis was excluded from the dashboard due to a large proportion of unidentifiable gender values, rendering cross-gender comparisons statistically unreliable.
- The dataset captures primarily English-language discourse. Non-English climate conversations, significant in Asia, Latin America, and Africa, are underrepresented.
- Bot and automated account activity has not been filtered. Future work should incorporate bot detection to ensure organic discourse is the subject of analysis.
- Regional classification uses simplified longitude banding. Country-level geocoding would yield stronger regional insights in a production context.